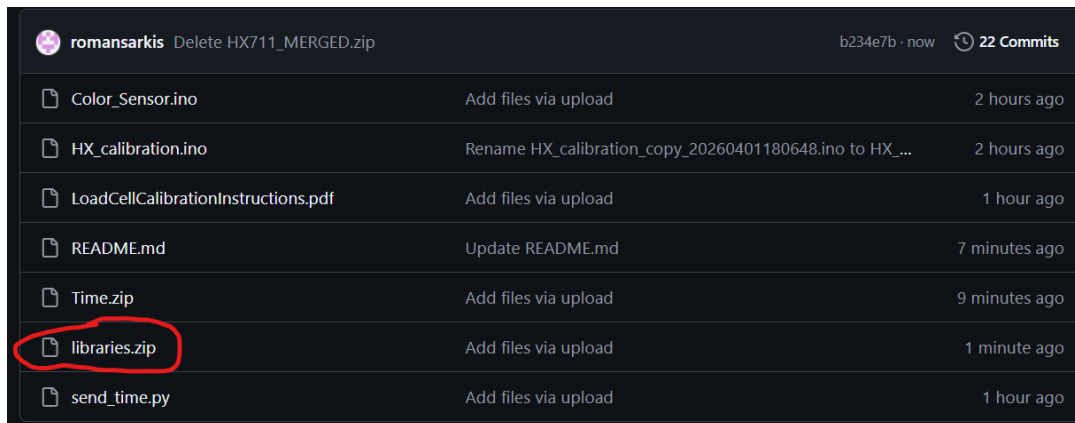


CSE 5911 Accurate Measurement of Bodily Fluid: Load Cell Calibration Instructions

The following document explains how to get an accurate calibration factor for each new load cell setup as you iterate on the physical prototype. Each new configuration/setup of the load cell within the physical device will likely require a new calibration factor to be calculated, so this step must be performed multiple times throughout the project's life cycle as you iterate.

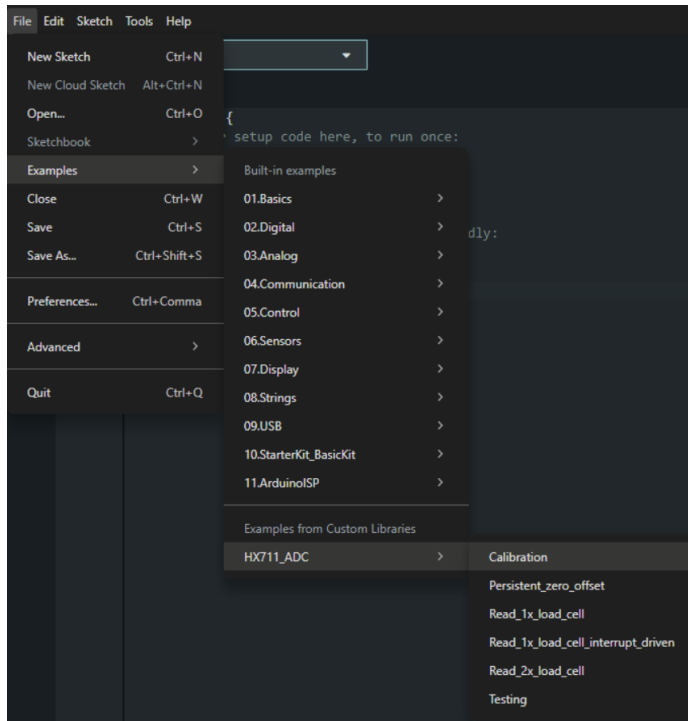
1. Download and install the Arduino IDE from their website: <https://www.arduino.cc/en/software/>
2. Download the Arduino library "HX711_ADC.zip" for calibrating the load cell with the Adafruit HX711 24-bit ADC for Load Cells / Strain Gauges from the project repo. It can be found in libraries.zip



 romansarkis	Delete HX711_MERGED.zip	b234e7b · now	🕒 22 Commits
 Color_Sensor.ino	Add files via upload	2 hours ago	
 HX_calibration.ino	Rename HX_calibration_copy_20260401180648.ino to HX_...	2 hours ago	
 LoadCellCalibrationInstructions.pdf	Add files via upload	1 hour ago	
 README.md	Update README.md	7 minutes ago	
 Time.zip	Add files via upload	9 minutes ago	
 libraries.zip	Add files via upload	1 minute ago	
 send_time.py	Add files via upload	1 hour ago	

3. Extract the downloaded zip. Then, locate the Arduino IDE's libraries folder. The path should look something like: C:\Users\Roman\Documents\Arduino\libraries. Each zip in the extracted libraries folder must also be individually extracted and put into the Arduino libraries folder. You cannot simply drag 'libraries' into your libraries folder.

4. Launch Arduino IDE and click File -> Examples. At the bottom, you should see HX711_ADC and example sketches from the newly imported custom library.



5. Then, choose Calibration. The example sketch should open in a new window.
6. 7. Change line 37 (`Serial.begin(57600); delay(10);`) to match your board's baud rate. In this case, change 57600 to 9600 since we are using an Arduino Mega2560.

```
Serial.begin(9600);
```

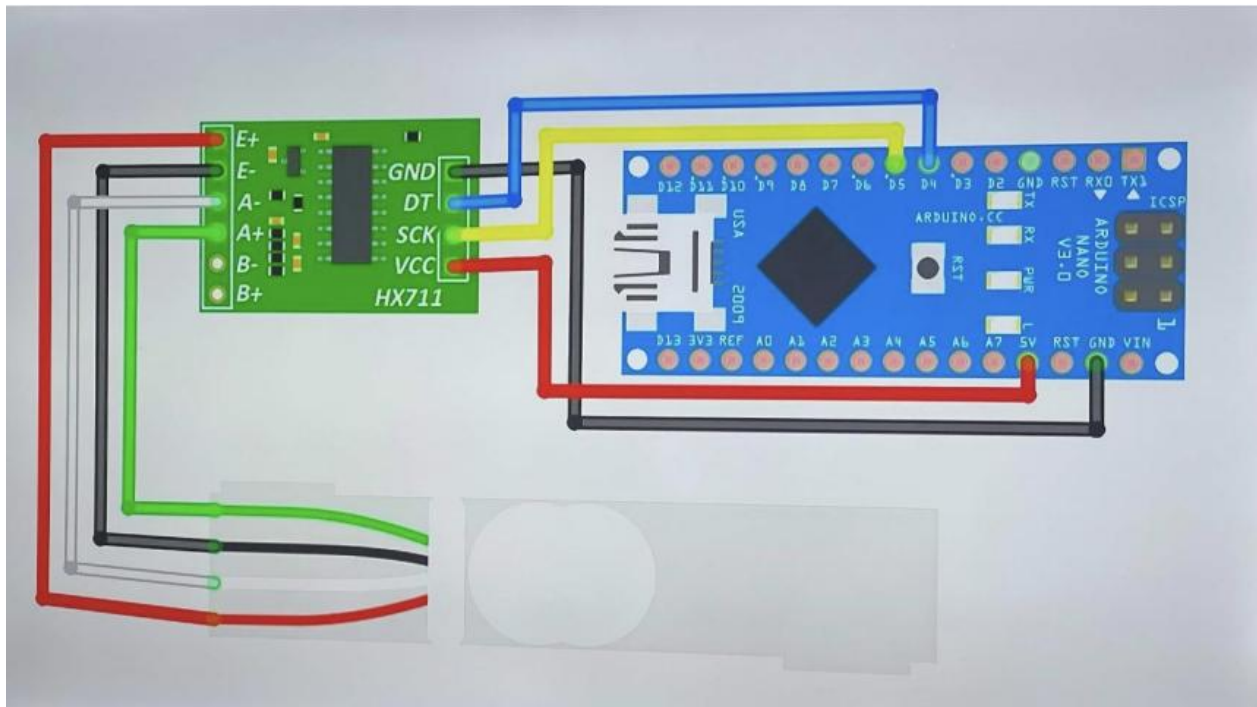
7. The pins numbers at the top of the file will also have to be adjusted depending on your setup. For our setup, we used pin 4 for data and pin 5 for the device's clock. This can be achieved by ensuring these lines match in calibration file:

```
uint8_t dataPin = 4;  
uint8_t clockPin = 5;
```

8. Connect the Arduino/prototype to your computer via a USB 2.0 type B cable, click upload (top left), and then click serial monitor (top right). The program should now be running on your Arduino and outputting legible data to the serial monitor. The calibration factor you get will be different across different load cells and prototype setups, so it will have to be recalculated for each new iteration of the physical prototype. If you're having issues here, consult the Arduino forums, you may be missing CH340 drivers for microcontrollers. <https://forum.arduino.cc/t/problems-installing-ch340-drivers-on-windows/1204813>

Useful references:

Pinout diagram for load cell sensor and HX711 load cell amplifier:



YouTube video from DIY Engineers:

<https://www.youtube.com/watch?v=pyeeyMsiNn0>